Class 12 Mathematics

Chapter 8: Application of Integrals

Practice Set B: step-wise solutions

20 questions **53** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ Board-pattern — 1 mark

The area of the region $\{(x, y) : |x| + |y| \leq 1\}$ is:

(A) 1 sq unit **(B)** 4 sq units **(C)** $\frac{1}{2}$ sq unit **(D)** 2 **sq units**

Formula / idea: Area under a line $y = f(x)$ from $x = a$ to $x = b$ is $\int_a^b y\,dx$.

1 In the first quadrant the boundary is the line $x + y = 1$

$y = 1 - x, \quad 0 \leq x \leq 1$

2 The region is symmetric about both axes: four equal parts

$$A = 4\int_0^1 (1 - x)\,dx$$

3 Integrate and put the limits; so the option is (D) ✓ 1

$$A = 4\left[x - \frac{x^2}{2}\right]_0^1 = 4 \times \frac{1}{2} = 2$$

Answer **(D)** 2 **sq units**

Watch out: Stopping at the first-quadrant part, $\frac{1}{2}$, option (C); the region has four equal parts.

2 MCQ Board-pattern 1 mark

If the area of the region in the first quadrant enclosed by the circle $x^2 + y^2 = r^2$ and the coordinate axes is 4π sq units, then r is equal to:

(A) 2 **(B)** 4 **(C)** 8 **(D)** 16

Formula / idea: $\displaystyle\int_0^r \sqrt{r^2 - x^2}\,dx = \left[\frac{x}{2}\sqrt{r^2 - x^2} + \frac{r^2}{2}\sin^{-1}\frac{x}{r}\right]_0^r = \frac{\pi r^2}{4}$

1 Area of the first-quadrant part

$$A = \int_0^r \sqrt{r^2 - x^2}\,dx = \frac{r^2}{2}\cdot\frac{\pi}{2} = \frac{\pi r^2}{4}$$

2 Equate to 4π and take $r > 0$; so the option is (B) ✓ 1

$$\frac{\pi r^2}{4} = 4\pi \Rightarrow r^2 = 16 \Rightarrow r = 4$$

Answer **(B)** 4

Watch out: Equating the whole circle, $\pi r^2 = 4\pi$, which gives $r = 2$, option (A).

3 MCQ Board-pattern 1 mark

If the area enclosed by the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{4} = 1\ (a > 0)$ is 10π sq units, then a is equal to:

(A) 25 **(B)** $\dfrac{5}{2}$ **(C)** 5 **(D)** 10

Formula / idea: Area enclosed by $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ is πab.

1 Here $b^2 = 4$, so $b = 2$

$$\text{Area} = \pi \cdot a \cdot 2 = 2\pi a$$

2 Equate to 10π; so the option is (C) ✓ 1

$$2\pi a = 10\pi \Rightarrow a = 5$$

Answer **(C)** 5

Watch out: Using $b^2 = 4$ in place of $b = 2$: $4\pi a = 10\pi$ gives $a = \dfrac{5}{2}$, option (B).

4 MCQ Board-pattern 1 mark

The area of the region bounded by the curve $y = \cos x$ and the x-axis between $x = 0$ and $x = \pi$ is:

(A) 2 **sq units** **(B)** 4 sq units **(C)** 3 sq units **(D)** 1 sq unit

Formula / idea: Area is always positive: take the modulus of the part below the x-axis.

1 $\cos x \geq 0$ on $\left[0, \frac{\pi}{2}\right]$ and $\cos x \leq 0$ on $\left[\frac{\pi}{2}, \pi\right]$, so split

$$A = \int_0^{\pi/2} \cos x \, dx + \left| \int_{\pi/2}^{\pi} \cos x \, dx \right|$$

2 Integrate and put the limits; so the option is (A) ✓ 1

$$A = [\sin x]_0^{\pi/2} + \left| [\sin x]_{\pi/2}^{\pi} \right| = 1 + |-1| = 2$$

Answer **(A)** 2 **sq units**

Watch out: Writing $\int_0^{\pi} \cos x \, dx = 0$ as the area; the part below the x-axis cancels the part above.

5 MCQ Board-pattern 1 mark

The area of the region bounded by the line $x = 2y + 3$, the y-axis and the lines $y = 1$ and $y = -1$ is:

(A) 4 sq units **(B)** $\frac{3}{2}$ sq units **(C)** 6 **sq units** **(D)** 8 sq units

Formula / idea: Area between a curve and the y-axis: $\int_c^d x \, dy$

1 For $-1 \leq y \leq 1$, $x = 2y + 3 > 0$: the region lies right of the y-axis

$$A = \int_{-1}^{1} (2y + 3) \, dy$$

2 Integrate and put the limits; so the option is (C) ✓ 1

$$A = \left[y^2 + 3y\right]_{-1}^{1} = (1 + 3) - (1 - 3) = 6$$

Answer **(C)** 6 **sq units**

Watch out: Writing the lower-limit value as $(1 + 3)$ instead of $(1 - 3)$, which gives 0.

6 **Assertion-Reason** Board-pattern 1 mark

Assertion (A): The area of the region bounded by the curve $y = \sqrt{x}$, the x-axis and the line $x = 4$ is $\frac{16}{3}$ sq units. **Reason (R):** The area of the region bounded by the curve $y = \sqrt{x}$, the y-axis and the line $y = 2$ is also $\frac{16}{3}$ sq units.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) **Assertion (A) is true, but Reason (R) is false.**

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: Area with the x-axis is $\int y\,dx$; area with the y-axis is $\int x\,dy$.

1 For A: vertical strips from $x = 0$ to $x = 4$

$$\int_0^4 \sqrt{x}\,dx = \frac{2}{3}\left[x^{3/2}\right]_0^4 = \frac{16}{3}$$

2 So A is true

3 For R: $x = y^2$; horizontal strips from $y = 0$ to $y = 2$

$$\int_0^2 y^2\,dy = \left[\frac{y^3}{3}\right]_0^2 = \frac{8}{3}$$

Check: $\frac{16}{3} + \frac{8}{3} = 8$, the 4×2 rectangle.

4 The area is $\frac{8}{3}$, not $\frac{16}{3}$, so R is false

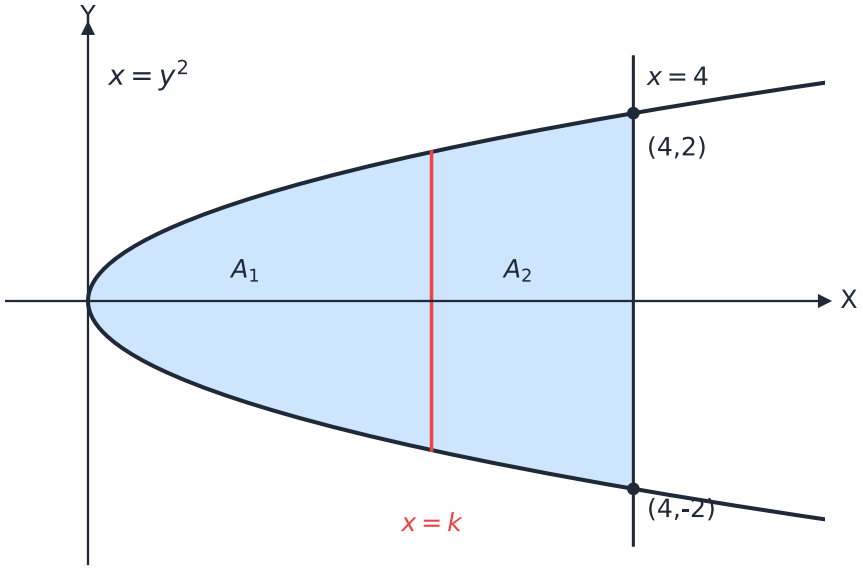
5 A is true and R is false, so the option is (C) ✓ 1

Answer **(C) A is true, but R is false**

Watch out: Assuming both regions are equal; together they fill the 4×2 rectangle, so they are $\frac{16}{3}$ and $\frac{8}{3}$.

7 **Short answer** PYQ-style 2 marks

The area between $x = y^2$ and $x = 4$ is divided into two equal parts by the line $x = k$. Find the value of k.



1	Total area, using symmetry about the x-axis (upper half $y = \sqrt{x}$) $A = 2\int_0^4 \sqrt{x}\,dx = \frac{4}{3}\left[x^{3/2}\right]_0^4 = \frac{32}{3}$	✓ ½
2	Area of the part from $x = 0$ to $x = k$ $A_1 = 2\int_0^k \sqrt{x}\,dx = \frac{4}{3}k^{3/2}$	✓ ½
3	Put $A_1 = \frac{1}{2}A$ $\frac{4}{3}k^{3/2} = \frac{16}{3} \Rightarrow k^{3/2} = 4$	✓ ½
4	Raise both sides to the power $\frac{2}{3}$ $k = 4^{2/3} = \sqrt[3]{16}$	✓ ½

Answer $k = 4^{2/3}$ **(that is,** $\sqrt[3]{16}$**)**

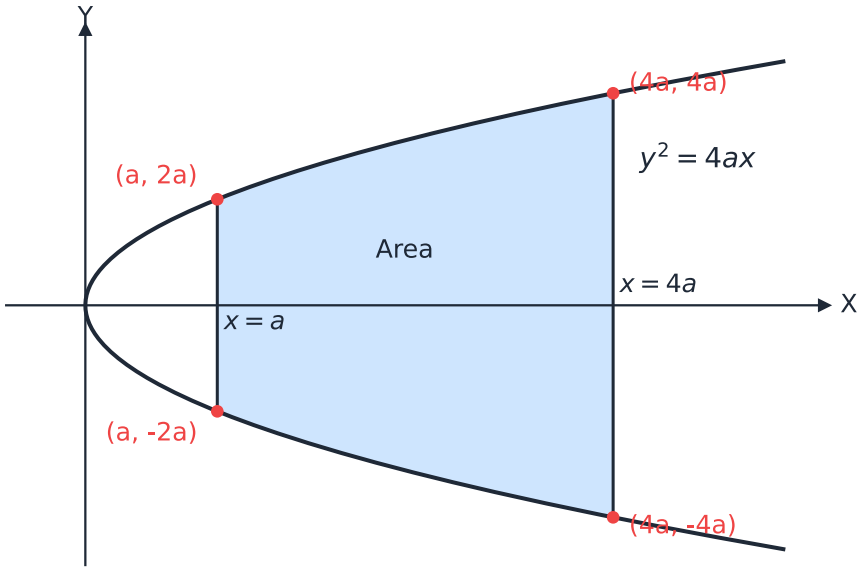
Watch out: Solving $k^{3/2} = 4$ as $k = 4^{3/2} = 8$; the power must be inverted, $k = 4^{2/3}$.

Level 2

18 marks

8 **Short answer** Board-pattern 3 marks

The curve $x = at^2, y = 2at\ (a > 0)$ is a parabola. Find the area of the region enclosed by this parabola and the ordinates (vertical lines) through its points where $t = 1$ and $t = 2$.



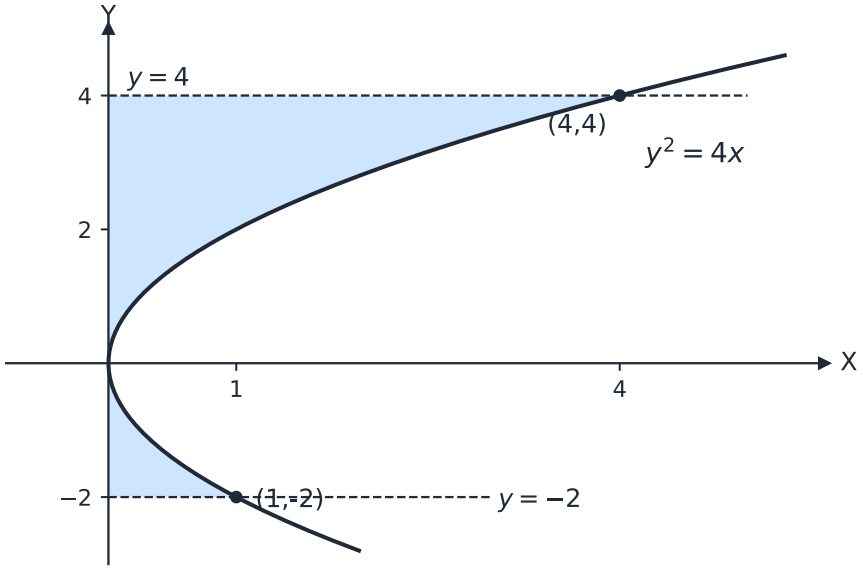
1	Eliminate t: put $t = \dfrac{y}{2a}$ in $x = at^2$ $x = a \cdot \dfrac{y^2}{4a^2} \Rightarrow y^2 = 4ax$	✓ ½
2	The ordinates: $t = 1$ gives $x = a$; $t = 2$ gives $x = 4a$	✓ ½
3	The region is symmetric about the x-axis; upper half $y = 2\sqrt{ax}$ $A = 2\displaystyle\int_a^{4a} 2\sqrt{a}\sqrt{x}\,dx$	✓ 1
4	Integrate $A = 4\sqrt{a} \cdot \dfrac{2}{3}\left[x^{3/2}\right]_a^{4a} = \dfrac{8\sqrt{a}}{3}\left(8a^{3/2} - a^{3/2}\right)$	✓ ½
5	Required area $A = \dfrac{8\sqrt{a}}{3} \cdot 7a^{3/2} = \dfrac{56a^2}{3}$ sq units	✓ ½

Answer $\dfrac{56a^2}{3}$ **sq units**

Watch out: Taking only the upper branch $y = 2\sqrt{ax}$ gives $\dfrac{28a^2}{3}$; the lines $x = a$, $x = 4a$ cut both branches.

9 **Short answer** Board-pattern 3 marks

Using integration, find the area of the region bounded by the parabola $y^2 = 4x$, the y-axis and the lines $y = -2$ and $y = 4$.



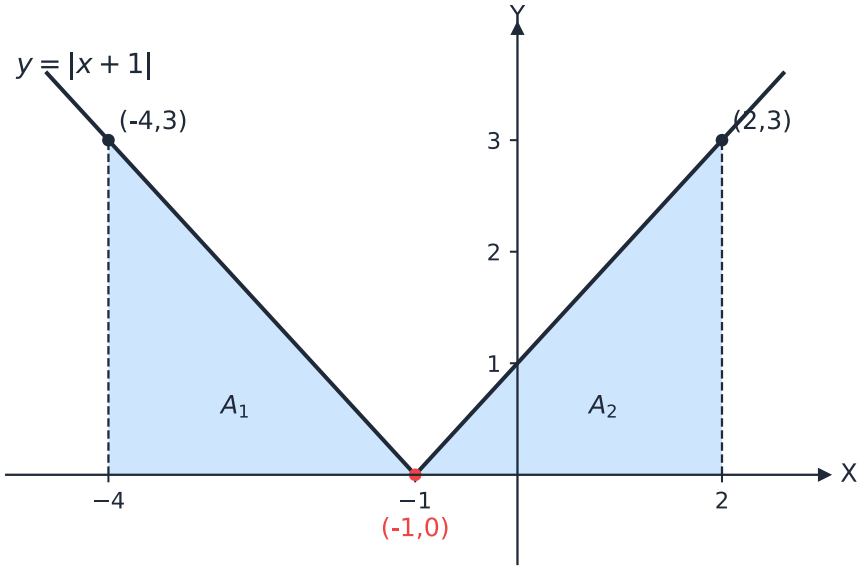
1	Sketch: the parabola meets $y = 4$ at $(4, 4)$ and $y = -2$ at $(1, -2)$	✓ ½
2	Write x in terms of y $y^2 = 4x \Rightarrow x = \dfrac{y^2}{4}$	
3	Take horizontal strips from $y = -2$ to $y = 4$ $A = \displaystyle\int_{-2}^{4} x\,dy = \frac{1}{4}\int_{-2}^{4} y^2\,dy$ $x = \dfrac{y^2}{4} \geq 0$, so no part needs a modulus.	✓ 1
4	Integrate and put the limits $A = \dfrac{1}{4}\left[\dfrac{y^3}{3}\right]_{-2}^{4} = \dfrac{1}{12}\left[64 - (-8)\right]$	✓ 1
5	Required area $A = \dfrac{72}{12} = 6$ sq units	✓ ½

Answer 6 **sq units**

Watch out: Writing $(-2)^3 = 8$, which gives $\dfrac{56}{12}$; the lower limit gives -8, so the bracket is $64 + 8$.

10 **Short answer** CBSE SQP 2025-26 3 marks

Sketch the graph of $y = |x + 1|$. Evaluate $\int_{-4}^{2} |x + 1|\, dx$. What does the value of this integral represent on the graph?



1	Sketch: a V-shape with vertex $(-1, 0)$, through $(-4, 3)$ and $(2, 3)$	✓ 1
2	Remove the modulus: $\|x + 1\| = -(x + 1)$ for $x < -1$ and $x + 1$ for $x \geq -1$ $\int_{-4}^{2} \|x + 1\|\, dx = \int_{-4}^{-1} -(x + 1)\, dx + \int_{-1}^{2} (x + 1)\, dx$	✓ ½
3	Integrate $= -\left[\frac{(x+1)^2}{2}\right]_{-4}^{-1} + \left[\frac{(x+1)^2}{2}\right]_{-1}^{2}$	✓ ½
4	Put the limits $= -\left(0 - \frac{9}{2}\right) + \left(\frac{9}{2} - 0\right) = 9$	✓ ½
5	It represents the area bounded by $y = \|x + 1\|$, the x-axis and the lines $x = -4$, $x = 2$	✓ ½

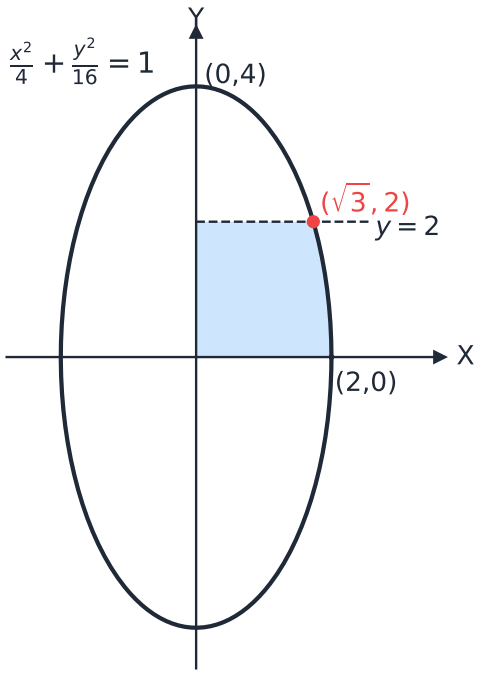
Answer **9; it is the area bounded by $y = |x + 1|$, the x-axis and the lines $x = -4$ and $x = 2$**

Watch out: Integrating $(x + 1)$ straight from -4 to 2 gives 0; the modulus must be removed piece by piece.

11 **Short answer** Board-pattern 2 marks

Using integration, find the area of the region in the first quadrant bounded by the ellipse $\frac{x^2}{4}+\frac{y^2}{16}=1$, the y-axis, the x-axis and the line $y=2$.

Formula / idea: $\int \sqrt{a^2-y^2}\,dy=\frac{y}{2}\sqrt{a^2-y^2}+\frac{a^2}{2}\sin^{-1}\frac{y}{a}+C$



1 Write x in terms of y (first quadrant) ✓ ½

$$\frac{x^2}{4}=\frac{16-y^2}{16}\Rightarrow x=\frac{1}{2}\sqrt{16-y^2}$$

2 Take horizontal strips from $y=0$ to $y=2$ ✓ ½

$$A=\int_0^2 x\,dy=\frac{1}{2}\int_0^2\sqrt{16-y^2}\,dy$$

3 Integrate and put the limits, using $\sin^{-1}\frac{1}{2}=\frac{\pi}{6}$ ✓ ½

$$A=\frac{1}{2}\left[\frac{y}{2}\sqrt{16-y^2}+8\sin^{-1}\frac{y}{4}\right]_0^2=\frac{1}{2}\left[\sqrt{12}+\frac{8\pi}{6}\right]$$

4 Required area ✓ ½

$$A=\frac{1}{2}\left[2\sqrt{3}+\frac{4\pi}{3}\right]=\left(\sqrt{3}+\frac{2\pi}{3}\right)\text{ sq units}$$

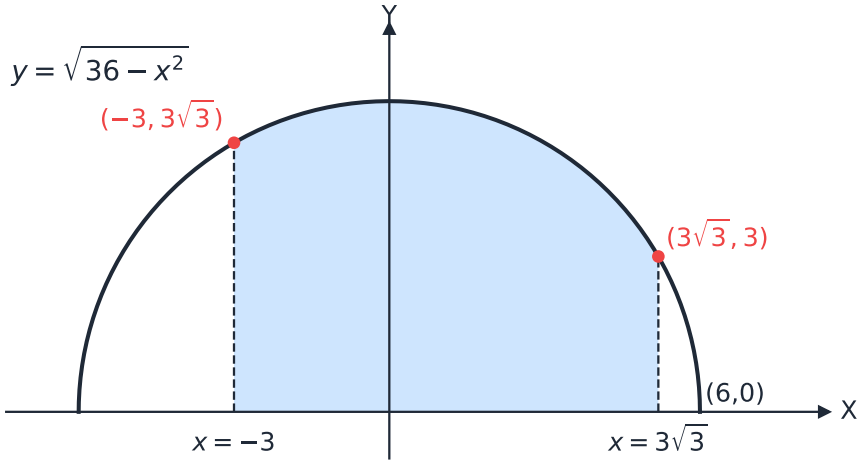
Answer $\left(\sqrt{3}+\frac{2\pi}{3}\right)$ **sq units**

Watch out: Writing $x=2\sqrt{16-y^2}$; from $\frac{x^2}{4}=\frac{16-y^2}{16}$ the factor is $\frac{1}{2}$.

12 **Short answer** Board-pattern 3 marks

Using integration, find the area of the region bounded by the curve $y = \sqrt{36 - x^2}$, the x-axis and the lines $x = -3$ and $x = 3\sqrt{3}$.

Formula / idea: $\displaystyle\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$



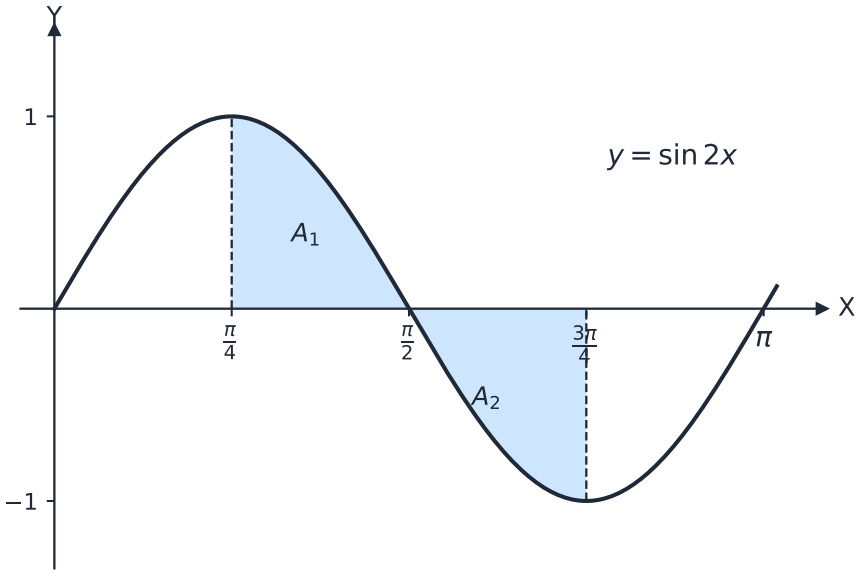
1	The curve is the upper half of $x^2 + y^2 = 36$; it lies above the x-axis	✓ ½
2	Write the area and use the formula for $\displaystyle\int \sqrt{a^2 - x^2}\,dx$ $A = \displaystyle\int_{-3}^{3\sqrt{3}} \sqrt{36 - x^2}\,dx = \left[\frac{x}{2}\sqrt{36 - x^2} + 18\sin^{-1}\frac{x}{6}\right]_{-3}^{3\sqrt{3}}$	✓ 1
3	Upper limit: $\sqrt{36 - 27} = 3$ and $\sin^{-1}\dfrac{\sqrt{3}}{2} = \dfrac{\pi}{3}$ $\dfrac{3\sqrt{3}}{2}\cdot 3 + 18\cdot\dfrac{\pi}{3} = \dfrac{9\sqrt{3}}{2} + 6\pi$	✓ ½
4	Lower limit: $\sqrt{36 - 9} = 3\sqrt{3}$ and $\sin^{-1}\left(-\dfrac{1}{2}\right) = -\dfrac{\pi}{6}$ $-\dfrac{3}{2}\cdot 3\sqrt{3} + 18\cdot\left(-\dfrac{\pi}{6}\right) = -\dfrac{9\sqrt{3}}{2} - 3\pi$	✓ ½
5	Subtract $A = \left(\dfrac{9\sqrt{3}}{2} + 6\pi\right) - \left(-\dfrac{9\sqrt{3}}{2} - 3\pi\right) = 9\sqrt{3} + 9\pi$ sq units	✓ ½

Answer $9\left(\sqrt{3} + \pi\right)$ **sq units**

Watch out: Taking $\sin^{-1}\left(-\dfrac{1}{2}\right) = \dfrac{\pi}{6}$; it is $-\dfrac{\pi}{6}$, and subtracting -3π adds 3π.

13 **Short answer** Board-pattern 2 marks

Find the area of the region bounded by the curve $y = \sin 2x$, the x-axis and the lines $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$.



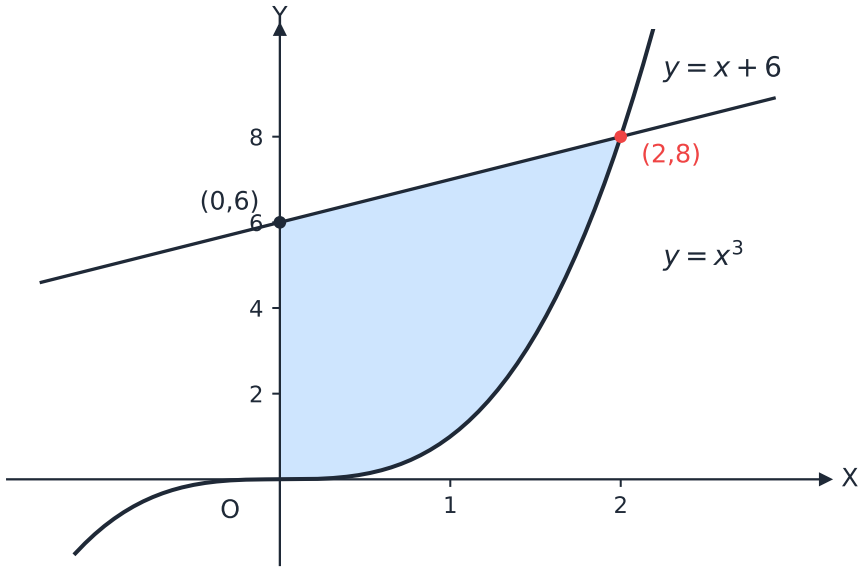
1	$\sin 2x = 0$ at $x = \frac{\pi}{2}$: curve above the x-axis before it, below after it	✓ ½
2	Split the area at $x = \frac{\pi}{2}$ $A = \int_{\pi/4}^{\pi/2} \sin 2x\, dx + \left\lvert \int_{\pi/2}^{3\pi/4} \sin 2x\, dx \right\rvert$	✓ ½
3	Integrate and put the limits $A = \left[-\frac{\cos 2x}{2}\right]_{\pi/4}^{\pi/2} + \left\lvert \left[-\frac{\cos 2x}{2}\right]_{\pi/2}^{3\pi/4} \right\rvert = \frac{1}{2} + \left\lvert -\frac{1}{2} \right\rvert$	✓ ½
4	Required area $A = 1$ sq unit	✓ ½

Answer 1 **sq unit**

Watch out: Integrating straight from $\frac{\pi}{4}$ to $\frac{3\pi}{4}$ gives 0, because the two equal parts cancel.

14 **Short answer** Board-pattern 2 marks

Find the area of the region bounded by the curve $y = x^3$, the line $y = x + 6$ and the line $x = 0$.



1	Meeting point of the curve and the line $x^3 = x + 6 \Rightarrow (x-2)(x^2+2x+3) = 0 \Rightarrow x = 2,\ \text{point}\ (2, 8)$ $x^2 + 2x + 3 = (x+1)^2 + 2 > 0$	✓ ½
2	For $0 \le x \le 2$ the line is above the curve $A = \int_0^2 \left[(x+6) - x^3\right] dx$	✓ ½
3	Integrate and put the limits $A = \left[\frac{x^2}{2} + 6x - \frac{x^4}{4}\right]_0^2 = 2 + 12 - 4$	✓ ½
4	Required area $A = 10$ sq units	✓ ½

Answer 10 **sq units**

Watch out: Subtracting the wrong way, $x^3 - (x+6)$, gives -10; check which graph is on top first.

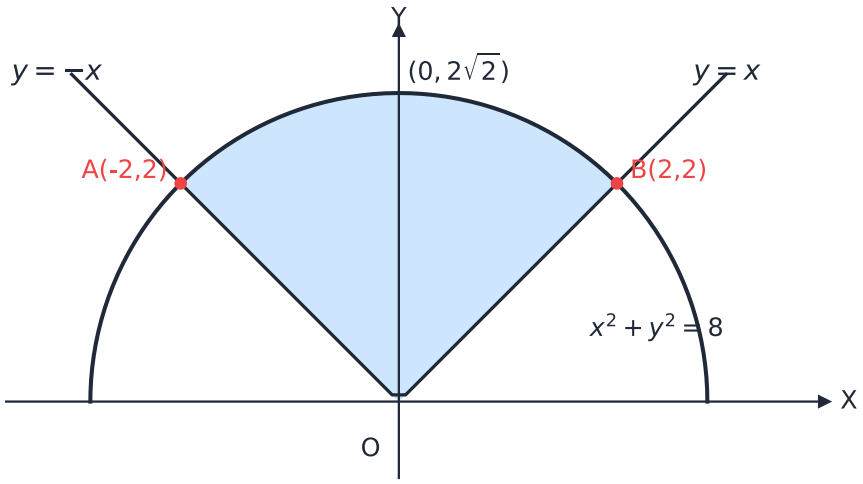
Level 3

27 marks

15 **Long answer** Board-pattern 5 marks

Using integration, find the area of the region $\{(x, y) : |x| \le y \le \sqrt{8 - x^2}\}$.

Formula / idea: $\int \sqrt{a^2 - x^2}\, dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$



1	$y = \sqrt{8 - x^2}$ is the upper half of $x^2 + y^2 = 8$ (centre O, radius $2\sqrt{2}$)	
2	$y \ge \|x\|$ means above both lines $y = x$ and $y = -x$	
3	Put $y = x$ in $x^2 + y^2 = 8$ for the corners $2x^2 = 8 \Rightarrow x = 2; \quad B(2, 2),$ and by symmetry $A(-2, 2)$	✓ 1
4	Sketch and shade the region between the two lines and the arc AB	✓ 1
5	The region is symmetric about the y-axis $A = 2\int_0^2 \left(\sqrt{8 - x^2} - x\right) dx$	✓ 1
6	Integrate $A = 2\left[\frac{x}{2}\sqrt{8 - x^2} + 4\sin^{-1}\frac{x}{2\sqrt{2}} - \frac{x^2}{2}\right]_0^2$	✓ 1
7	Put the limits, using $\sin^{-1}\frac{1}{\sqrt{2}} = \frac{\pi}{4}$ $A = 2\left[1 \cdot 2 + 4 \cdot \frac{\pi}{4} - 2\right] - 0$	✓ ½

8 Required area

$A = 2\pi$ sq units

Check: a quarter-circle sector, $\frac{1}{4}\pi(2\sqrt{2})^2 = 2\pi$.

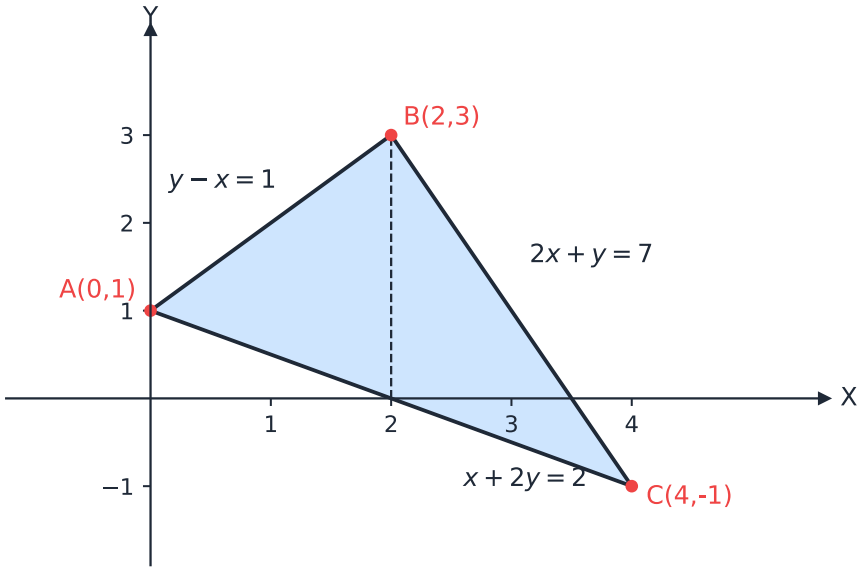
✓ ½

Answer 2π **sq units**

Watch out: Writing the $\sin^{-1}$ coefficient as 8: in $\frac{a^2}{2}\sin^{-1}\frac{x}{a}$, $a^2 = 8$, so it is 4.

16 **Long answer** Board-pattern 5 marks

Using integration, compute the area of the region bounded by the lines $x + 2y = 2$, $y - x = 1$ and $2x + y = 7$.



1 Solve the lines in pairs to get the vertices

$y - x = 1,\ x + 2y = 2 \Rightarrow A(0, 1)$

$y - x = 1,\ 2x + y = 7 \Rightarrow B(2, 3)$

$x + 2y = 2,\ 2x + y = 7 \Rightarrow C(4, -1)$

✓ 1

2 Sketch the triangle ABC

✓ ½

3 Top side: AB on $[0, 2]$, BC on $[2, 4]$; bottom side: AC on $[0, 4]$

$AB : y = x + 1, \quad BC : y = 7 - 2x, \quad AC : y = \frac{2 - x}{2}$

✓ ½

4 Area = (upper − lower) on each part

$$A = \int_0^2 \left[(x + 1) - \frac{2 - x}{2}\right] dx + \int_2^4 \left[(7 - 2x) - \frac{2 - x}{2}\right] dx$$

Upper minus lower works even below the x-axis.

✓ 1

5 First integral (the integrand simplifies to $\frac{3x}{2}$) ✓ ½

$$\int_0^2 \frac{3x}{2}\,dx = \left[\frac{3x^2}{4}\right]_0^2 = 3$$

6 Second integral (the integrand simplifies to $\frac{12-3x}{2}$) ✓ ½

$$\frac{1}{2}\left[12x - \frac{3x^2}{2}\right]_2^4 = \frac{1}{2}\left[(48-24)-(24-6)\right] = 3$$

7 Required area ✓ 1

$$A = 3 + 3 = 6 \text{ sq units}$$

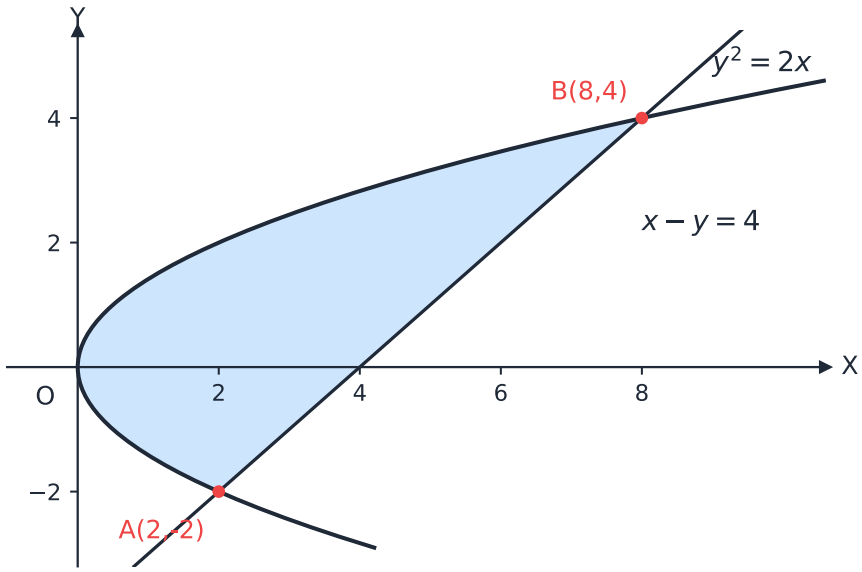
Answer 6 **sq units**

Watch out: Treating $\int_0^4 \frac{2-x}{2}\,dx = 0$ as the area under AC; AC crosses the x-axis, so use (top – bottom).

17 **Long answer** Board-pattern 5 marks

Find the area of the region bounded by the parabola $y^2 = 2x$ and the straight line $x - y = 4$.

Formula / idea: With horizontal strips, area $= \int_c^d (x_{\text{right}} - x_{\text{left}})\,dy$.



1 Put $x = y + 4$ in $y^2 = 2x$ ✓ ½

$$y^2 = 2y + 8 \Rightarrow (y-4)(y+2) = 0 \Rightarrow y = 4,\ -2$$

2 So the curves meet at $A(2,-2)$ and $B(8,4)$ ✓ ½

3 Sketch: the region lies between the parabola (left) and the line (right) ✓ 1

4 Take horizontal strips: for $-2 \le y \le 4$, $x_{\text{line}} = y + 4 \ge x_{\text{parabola}} = \dfrac{y^2}{2}$

$$A = \int_{-2}^{4} \left[(y+4) - \frac{y^2}{2} \right] dy$$

Vertical strips would need two integrals, split at $x = 2$. ✓ 1

5 Integrate

$$A = \left[\frac{y^2}{2} + 4y - \frac{y^3}{6} \right]_{-2}^{4}$$

✓ ½

6 Put the limits

$$A = \left(8 + 16 - \frac{32}{3} \right) - \left(2 - 8 + \frac{4}{3} \right) = \frac{40}{3} + \frac{14}{3}$$

✓ 1

7 Required area

$$A = \frac{54}{3} = 18 \text{ sq units}$$

✓ ½

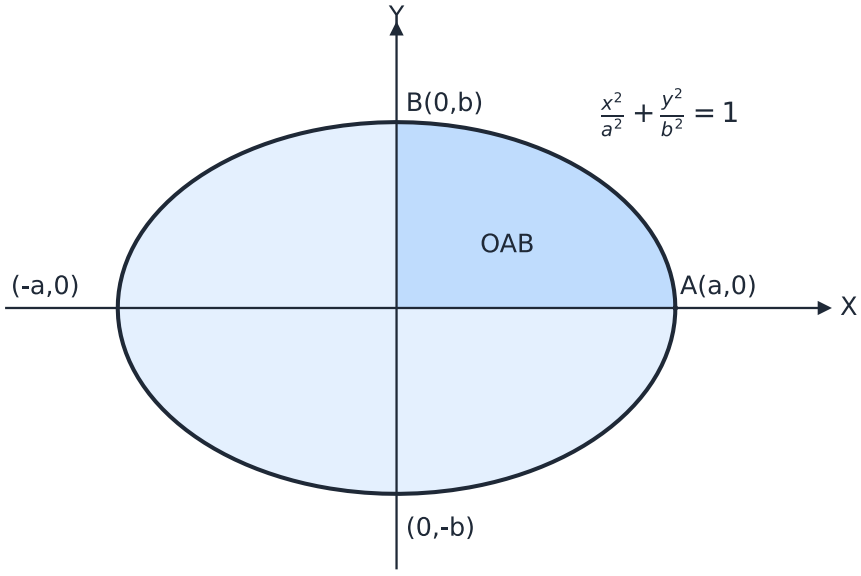
Answer 18 **sq units**

Watch out: Using vertical strips with $\sqrt{2x} - (x-4)$ from 0 to 8: for $0 \le x \le 2$ the lower boundary is $-\sqrt{2x}$, not the line.

18 **Long answer** PYQ-style 5 marks

Using integration, find the area enclosed by the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$. Hence, find the area enclosed by the ellipse $16x^2 + 9y^2 = 144$.

Formula / idea: $\displaystyle\int \sqrt{a^2 - x^2}\, dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$



1 Sketch the ellipse through $(\pm a, 0)$ and $(0, \pm b)$ ✓ ½

2 In the first quadrant ✓ ½

$$y = \frac{b}{a}\sqrt{a^2 - x^2}$$

3 The ellipse is symmetric about both axes: area $= 4\times$ area OAB ✓ 1

$$A = 4\int_0^a \frac{b}{a}\sqrt{a^2 - x^2}\,dx$$

4 Integrate ✓ 1

$$A = \frac{4b}{a}\left[\frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}\right]_0^a$$

5 Put the limits ✓ ½

$$A = \frac{4b}{a}\left[0 + \frac{a^2}{2}\cdot\frac{\pi}{2} - 0\right] = \pi ab \text{ sq units}$$

6 Hence: divide $16x^2 + 9y^2 = 144$ by 144 ✓ ½

$$\frac{x^2}{9} + \frac{y^2}{16} = 1 \Rightarrow a = 3,\ b = 4$$

7 Required area ✓ 1

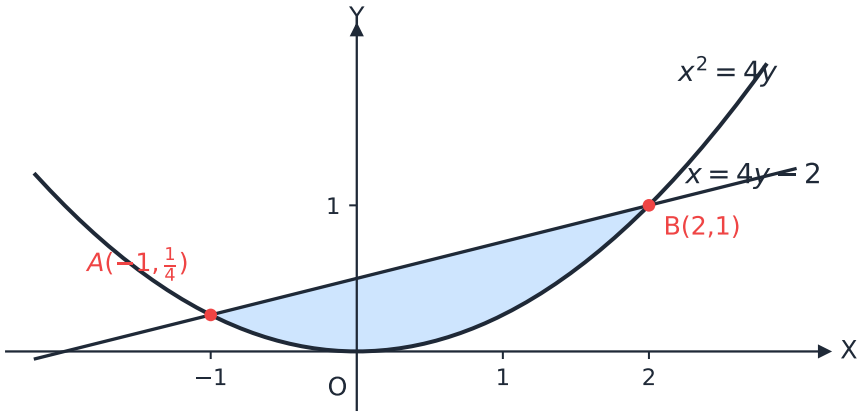
$$A = \pi \times 3 \times 4 = 12\pi \text{ sq units}$$

Answer πab **sq units;** 12π **sq units**

Watch out: Putting $a^2 = 9$ and $b^2 = 16$ into πab to get 144π; use the semi-axes $a = 3, b = 4$.

19 **Short answer** Board-pattern 3 marks

Using integration, find the area of the region bounded by the curve $x^2 = 4y$ and the straight line $x = 4y - 2$.



1 Put $4y = x + 2$ in $x^2 = 4y$ ✓ 1

$$x^2 = x + 2 \Rightarrow (x - 2)(x + 1) = 0; \quad A\left(-1, \tfrac{1}{4}\right),\ B(2, 1)$$

2 For $-1 \le x \le 2$ the line $y = \dfrac{x+2}{4}$ lies above the parabola $y = \dfrac{x^2}{4}$

$$A = \int_{-1}^{2} \left[\frac{x+2}{4} - \frac{x^2}{4}\right] dx$$

✓ 1

3 Integrate and put the limits

$$A = \frac{1}{4}\left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2} = \frac{1}{4}\left[\left(2 + 4 - \frac{8}{3}\right) - \left(\frac{1}{2} - 2 + \frac{1}{3}\right)\right]$$

✓ ½

4 Required area

$$A = \frac{1}{4}\left[\frac{10}{3} + \frac{7}{6}\right] = \frac{1}{4} \cdot \frac{27}{6} = \frac{9}{8} \text{ sq units}$$

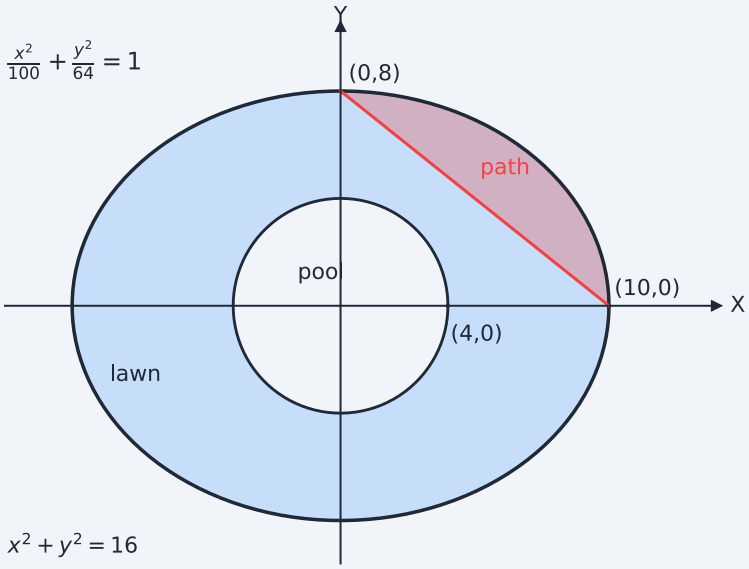
✓ ½

Answer $\dfrac{9}{8}$ **sq units**

Watch out: At $x = -1$, writing $-\dfrac{x^3}{3} = -\dfrac{1}{3}$; since $(-1)^3 = -1$, it is $+\dfrac{1}{3}$.

20 Case study Board-pattern 4 marks

Case Study: A housing society has an elliptical lawn whose boundary is the ellipse $\dfrac{x^2}{100} + \dfrac{y^2}{64} = 1$, with its centre at the origin (units in metres). At the centre there is a circular fountain pool with boundary $x^2 + y^2 = 16$. Grass covers the lawn everywhere outside the pool. A straight footpath of negligible width joins the points $(10, 0)$ and $(0, 8)$ on the boundary. Based on this information, answer the following questions using integration.



(i) Find the area enclosed by the fountain pool. 1 mark

(ii) Find the area enclosed by the elliptical boundary of the lawn. 1 mark

(iii)(a) Find the area of the part of the lawn in the first quadrant that lies between the footpath and the boundary. 2 marks

OR

(iii)(b) Find the area of the grassed part of the lawn lying in the first quadrant. 2 marks

Formula / idea: $\displaystyle\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$

(i)

1 mark

1 Pool radius $= 4$; the circle is symmetric about both axes ✓ ½

$$A = 4\int_0^4 \sqrt{16-x^2}\,dx = 4\left[\frac{x}{2}\sqrt{16-x^2} + 8\sin^{-1}\frac{x}{4}\right]_0^4$$

2 Put the limits ✓ ½

$$A = 4 \times 8 \cdot \frac{\pi}{2} = 16\pi \text{ m}^2$$

Answer (i) 16π m^2

(ii)

1 mark

1 First quadrant: $y = \frac{4}{5}\sqrt{100-x^2}$; use symmetry about both axes ✓ ½

$$A = 4\int_0^{10} \frac{4}{5}\sqrt{100-x^2}\,dx = \frac{16}{5}\left[\frac{x}{2}\sqrt{100-x^2} + 50\sin^{-1}\frac{x}{10}\right]_0^{10}$$

2 Put the limits ✓ ½

$$A = \frac{16}{5} \times 50 \cdot \frac{\pi}{2} = 80\pi \text{ m}^2$$

Answer (ii) 80π m^2

(iii)(a)

2 marks

1 Equation of the footpath (intercept form) ✓ ½

$$\frac{x}{10} + \frac{y}{8} = 1 \Rightarrow y = \frac{4}{5}(10-x)$$

2 Boundary on top, path below, for $0 \le x \le 10$ ✓ ½

$$A = \int_0^{10} \left[\frac{4}{5}\sqrt{100-x^2} - \frac{4}{5}(10-x)\right] dx$$

3 First part is $\frac{1}{4}$ of (ii); integrate the second ✓ ½

$$A = 20\pi - \frac{4}{5}\left[10x - \frac{x^2}{2}\right]_0^{10} = 20\pi - \frac{4}{5} \times 50$$

4 Required area ✓ ½

$$A = (20\pi - 40) \text{ m}^2$$

Answer (iii)(a) $(20\pi - 40)$ m^2

OR

(iii)(b)

2 marks

1 Lawn in the first quadrant (a quarter of the ellipse) ✓ 1

$$A_1 = \int_0^{10} \frac{4}{5}\sqrt{100-x^2}\,dx = \frac{4}{5} \times 50 \cdot \frac{\pi}{2} = 20\pi$$

2	Pool in the first quadrant $A_2 = \int_0^4 \sqrt{16 - x^2}\, dx = 8 \cdot \frac{\pi}{2} = 4\pi$	✓ ½
3	Grassed part = lawn − pool $A = 20\pi - 4\pi = 16\pi \text{ m}^2$	✓ ½

Answer (iii)(b) $16\pi \text{ m}^2$

Watch out: In (iii)(b), forgetting to remove the pool: the grassed part is the quarter-ellipse minus the quarter-circle.